KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET, DAN TEKNOLOGI
**POLITEKNIK NEGERI MALANG**
**JURUSAN TEKNOLOGI INFORMASI**
Jl. Soekarno Hatta No. 9, Jatimulyo, Lowokwaru, Malang 65141
Telp. (0341) 404424 – 404425, Fax (0341) 404420
https://jti.polinema.ac.id

| | | |
|---|---|---|
| Mata Kuliah | : | Pemograman Framework |
| Program Studi | : | D4 – Teknik Informatika |
| Semester | : | 6 |

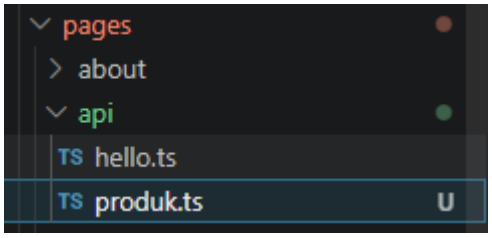
| | | |
|---|---|---|
| Kelas | : | TI – 3B |
| NIM | : | 2341720169 |
| Nama | : | Dimas Adi Bayu S |
| Jobsheet ke- | : | 7 |

# Laporan Jobsheet

## Praktikum 1

| Langkah | Jawaban/Deskripsi |
|---|---|
| 1 | Menjalankan Project dengan npm run dev |

## Praktikum Ke-2

| Langkah | Jawaban/Deskripsi |
|---|---|
| 1 | Buat file pad pages/api/produk.js   |
| 2 | Tambahkan data statis |

```
import type { NextApiRequest, NextApiResponse } from "next";

Windsurf: Refactor | Explain
type Data = {
    status: boolean;
    status_code : number;

    data:{
        id: string;
        nama: string;
        harga: number;
        ukuran: string;
        warna: string;
    }[];
};
```

```
export default function handler(
    req: NextApiRequest,
    res: NextApiResponse<Data>
)
    {
        const data =
        [
            {
                id: "1",
                nama: "Kaos polos",
                harga: 100000,
                ukuran: "L",
                warna: "Hitam",
            },
            {
                id: "2",
                nama: "kaos lengan panjang",
                harga: 300000,
                ukuran: "XL",
                warna: "Merah",
            },
            {
                id: "3",
                nama: "Sepatu",
                harga: 250000,
                ukuran: "43",
                warna: "Hitam",
            },
        ];
        res.status(200).json({status: true,status_code: 200,data:data });
    }
```

| | |
|---|---|
| 3 | Akses: http://localhost:3000/api/produk |

| | |
|---|---|
| | http://localhost:3000/api/produk<br>Pretty-print<br>{"status":true,"status_code":200,"data":[{"id":"1","nama":"Kaos polos","harga":100000,"ukuran":"L","warna":"Hitam"},{"id":"2","nama":"kaos lengan panjang","harga":300000,"ukuran":"XL","warna":"Merah"},{"id":"3","nama":"Sepatu","harga":250000,"ukuran":"43","warna":"Hitam"}]} |

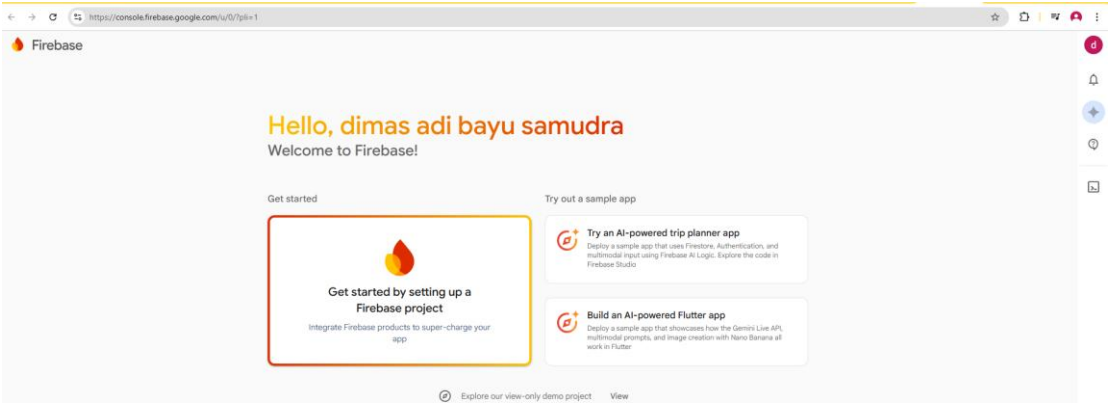
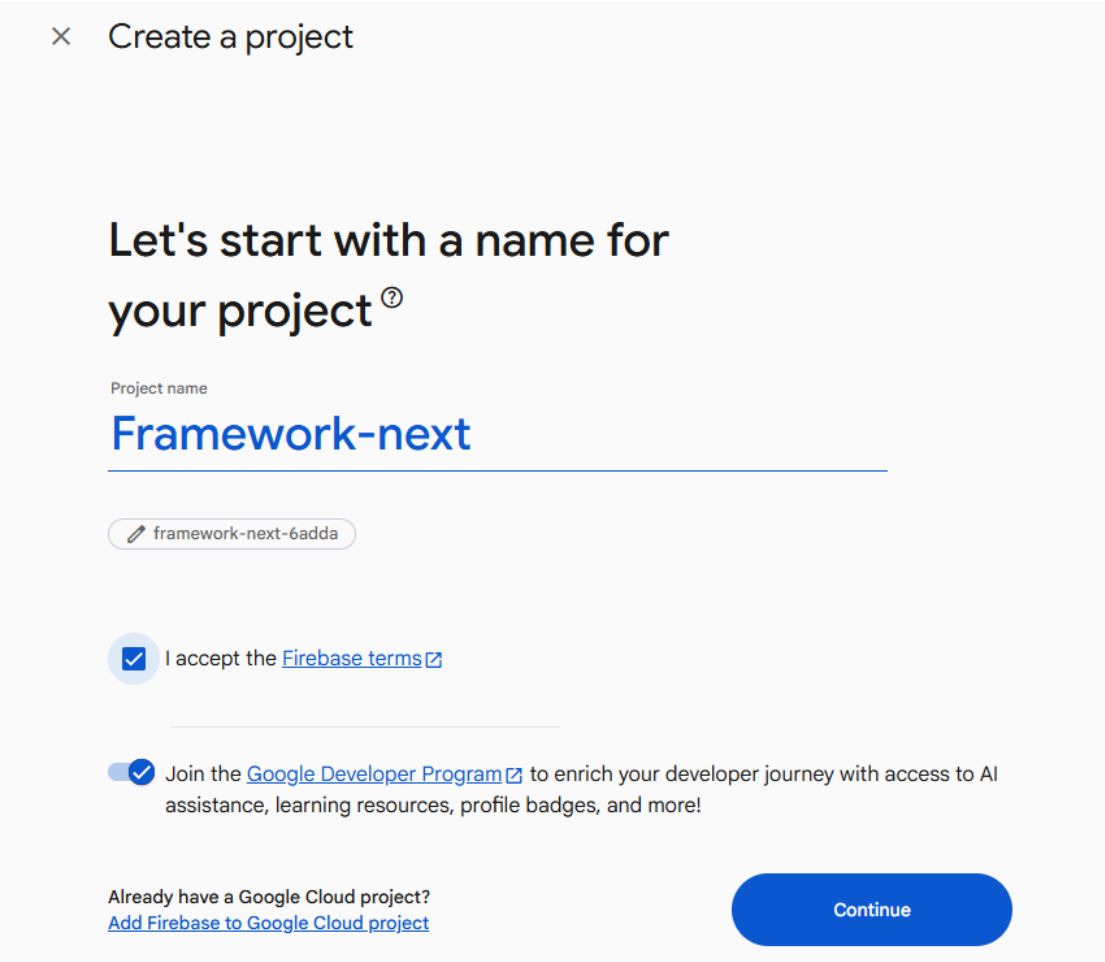
## Praktikum Ke-3

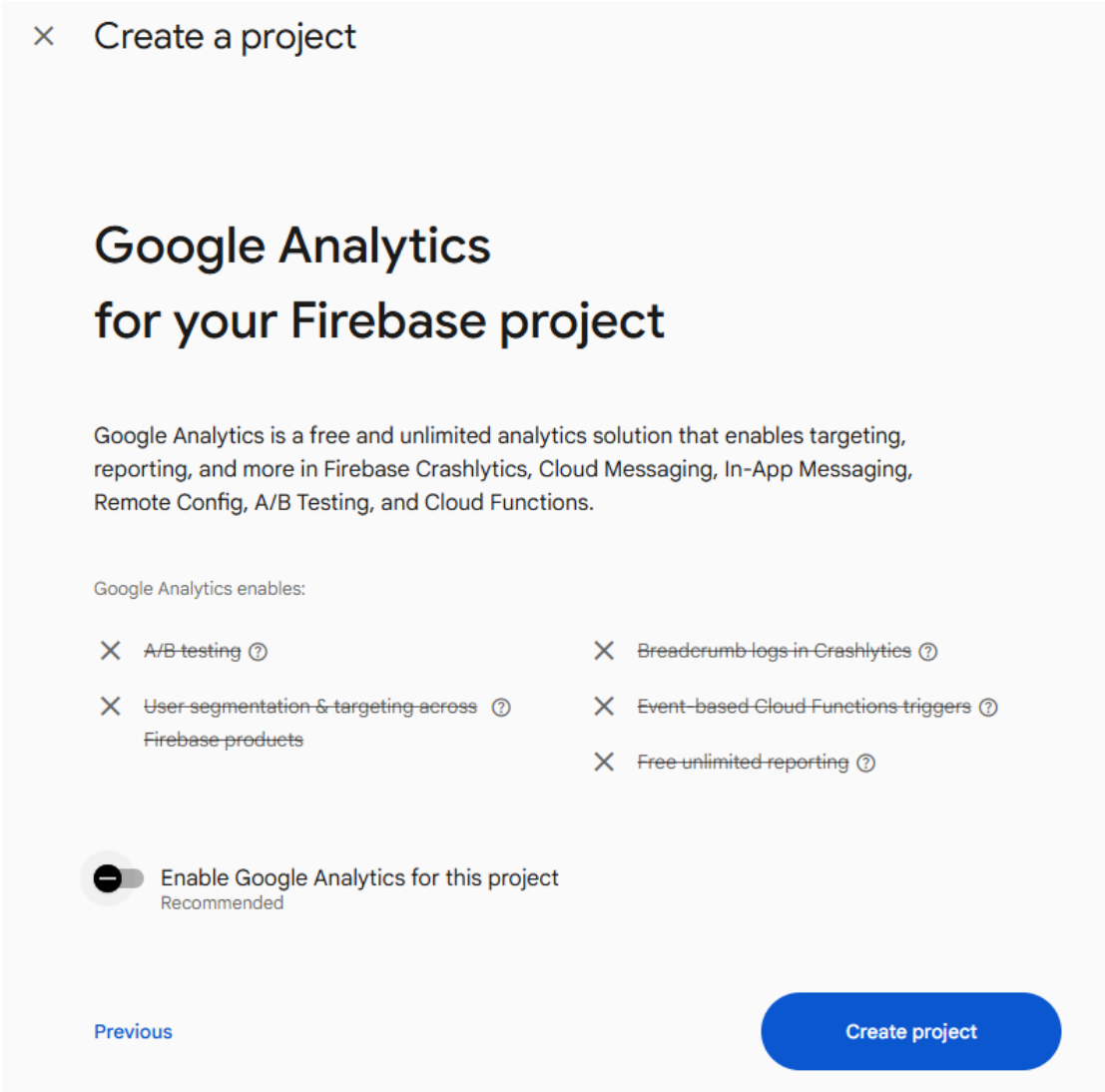
| Langkah | Jawaban/Deskripsi |
|---|---|
| 1 | Buka pages/product/index.tsx<br>• Modifikasi kode<br>o Tambahkan useEffect() dan comment useEffect untuk isLogin |

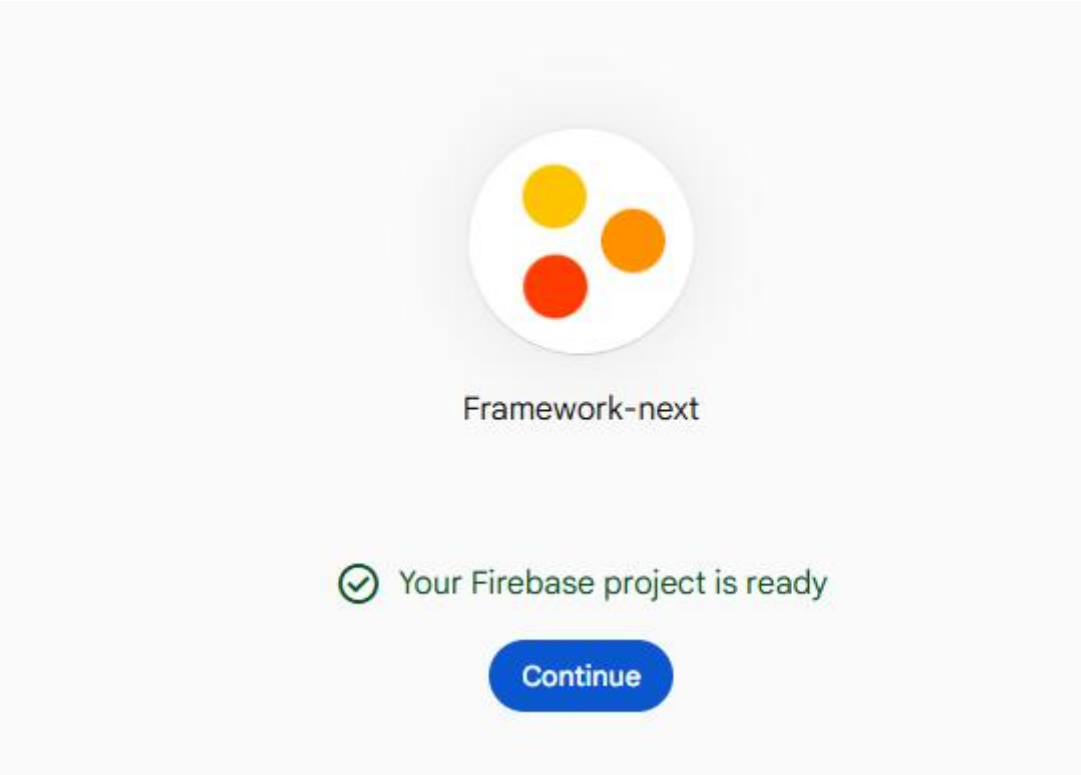
```
import ProdukPage from "../../views/produk";
import { useEffect, useState } from "react";

Windsurf: Refactor | Explain
type ProductType = {
    id: string;
    nama: string;
    harga: number;
    ukuran: string;
    warna: string;
};

Windsurf: Refactor | Explain | Generate JSDoc | X
const kategori = () => {
    const [products, setProducts] = useState<ProductType[]>([]);

    useEffect(() => {
  fetch("/api/produk")
    .then((response) => response.json())
    .then((responsedata) => {
      //console.log("Data produk:", responsedata.data);
      setProducts(responsedata.data);
    })
    .catch((error) => {
      console.error("Error fetching produk:", error);
    });
}, []);
```

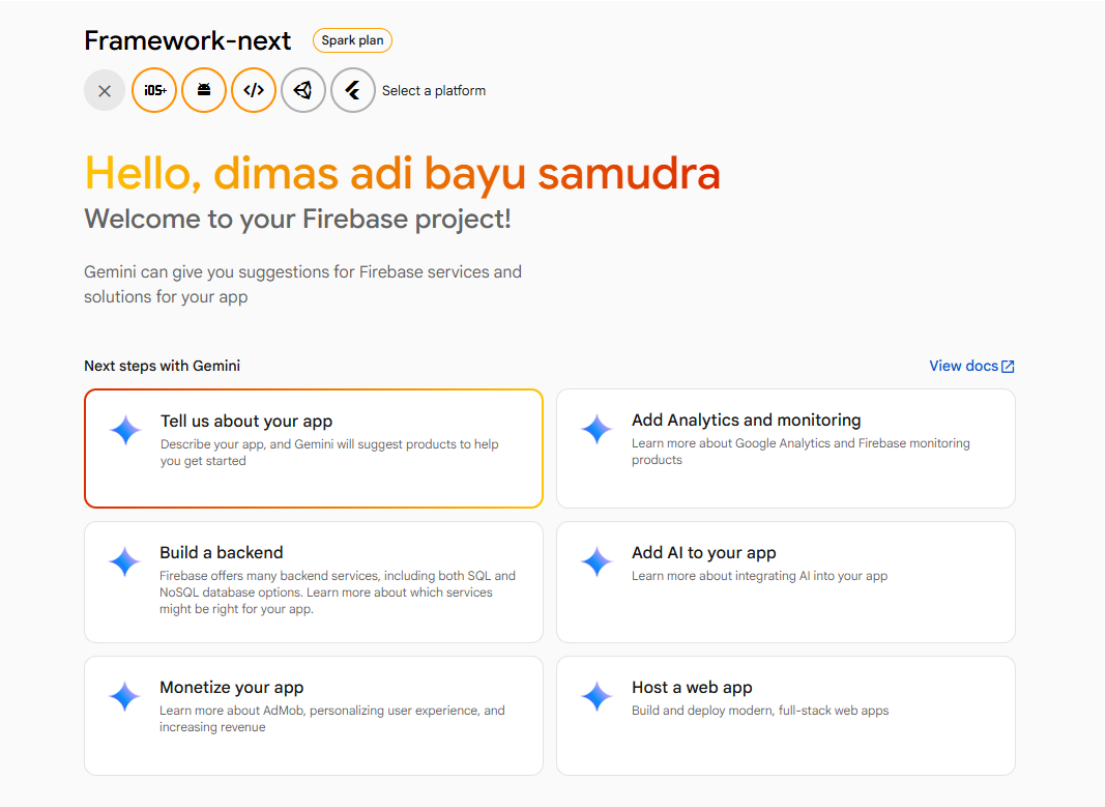
| | |
|---|---|
| | ```<br>return (  You, 1 second ago • Uncommitted<br>  <div><br>  <h1>Daftar Produk</h1><br>  {products.map((product: ProductType) => (<br>  <div key={product.id}><br>  <h2>{product.nama}</h2><br>  <p>Harga: {product.harga}</p><br>  <p>Ukuran: {product.ukuran}</p><br>  <p>Warna: {product.warna}</p><br>  </div><br>  ))}<br>  </div><br>);<br>};<br><br>export default kategori;<br>```<br>(dark theme) |
| 2 | Jalankan browser http://localhost:3000/produk<br> <br> |

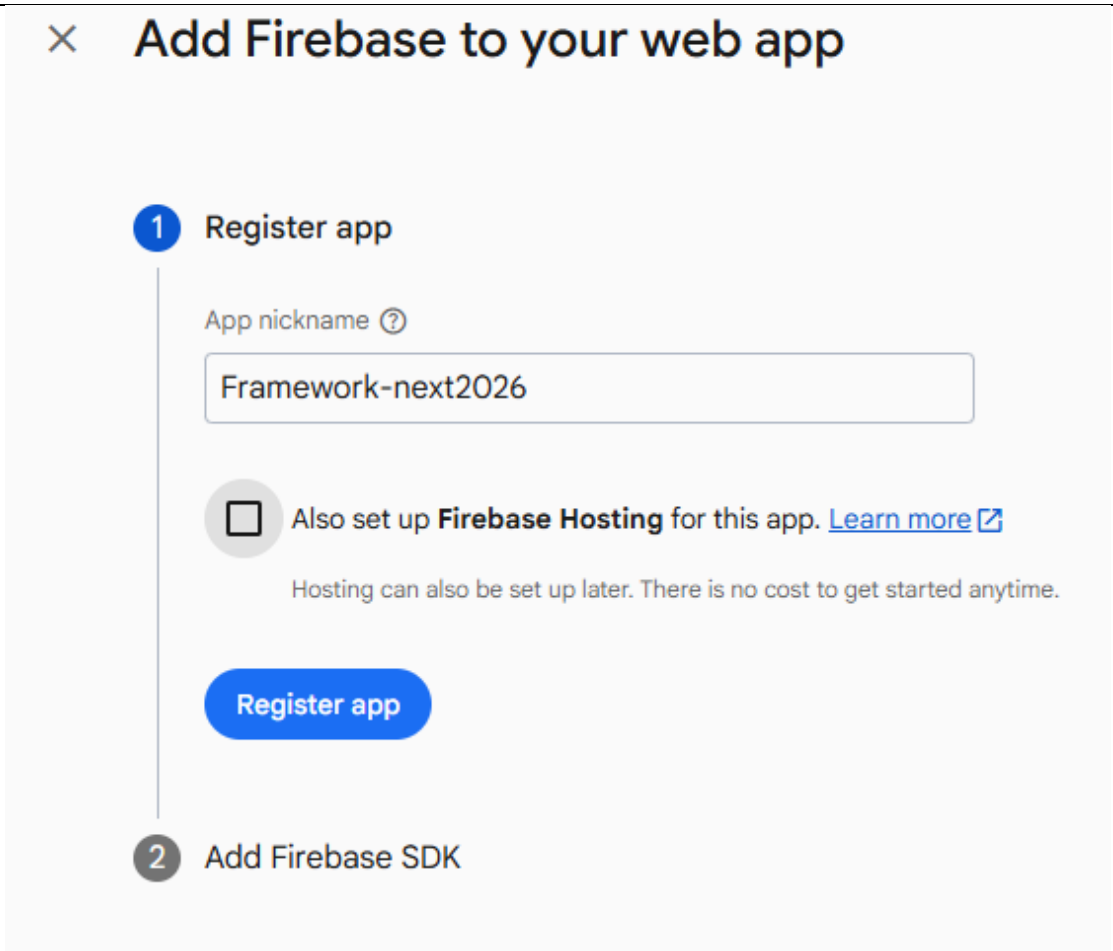
**Praktikum Ke-4**

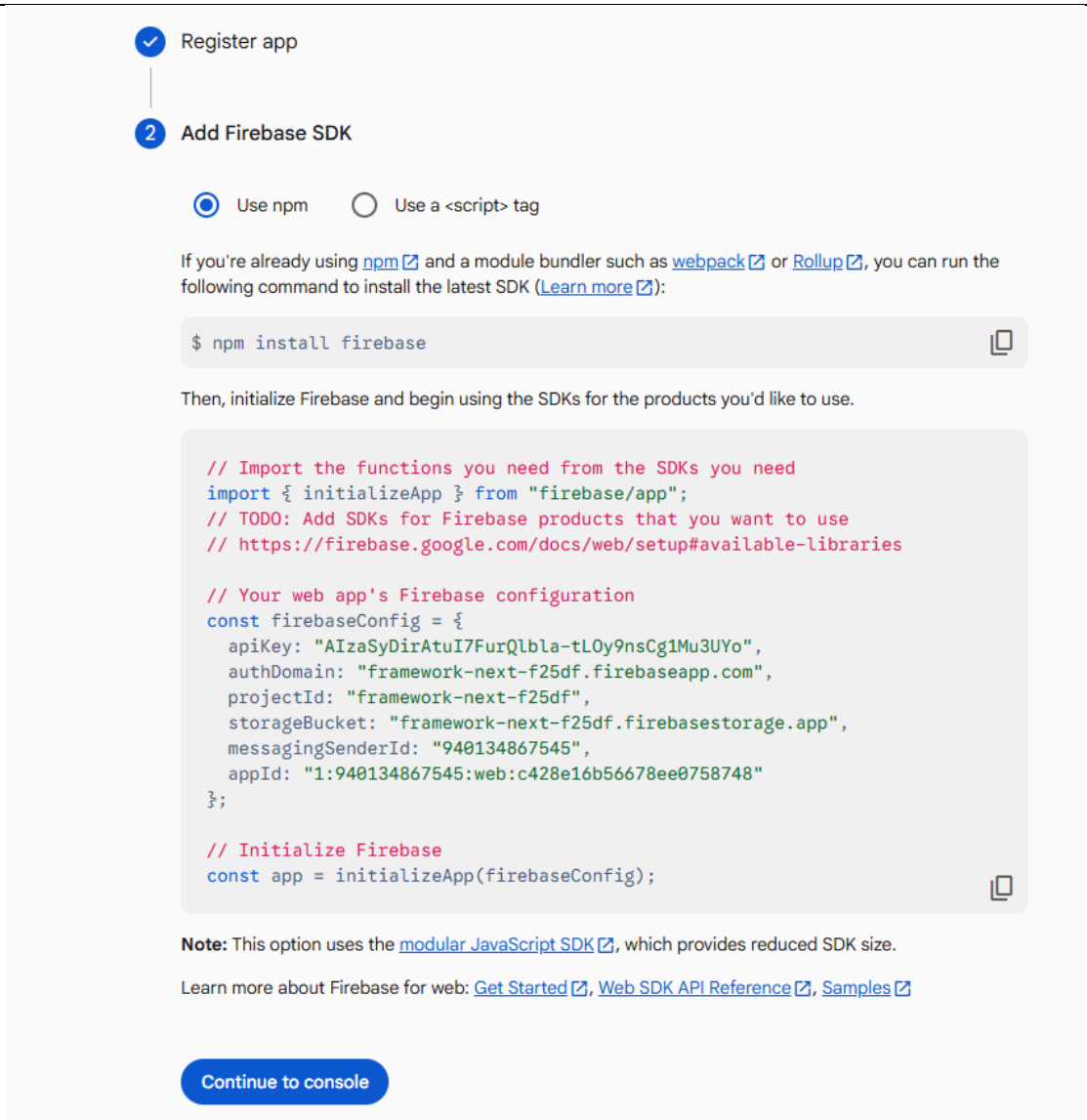
| Langkah | Jawaban/Deskripsi |
|---|---|
| 1 | Buka Firebase Go To Console ( login dengan login google)   |
| 2 | Buat project baru   |

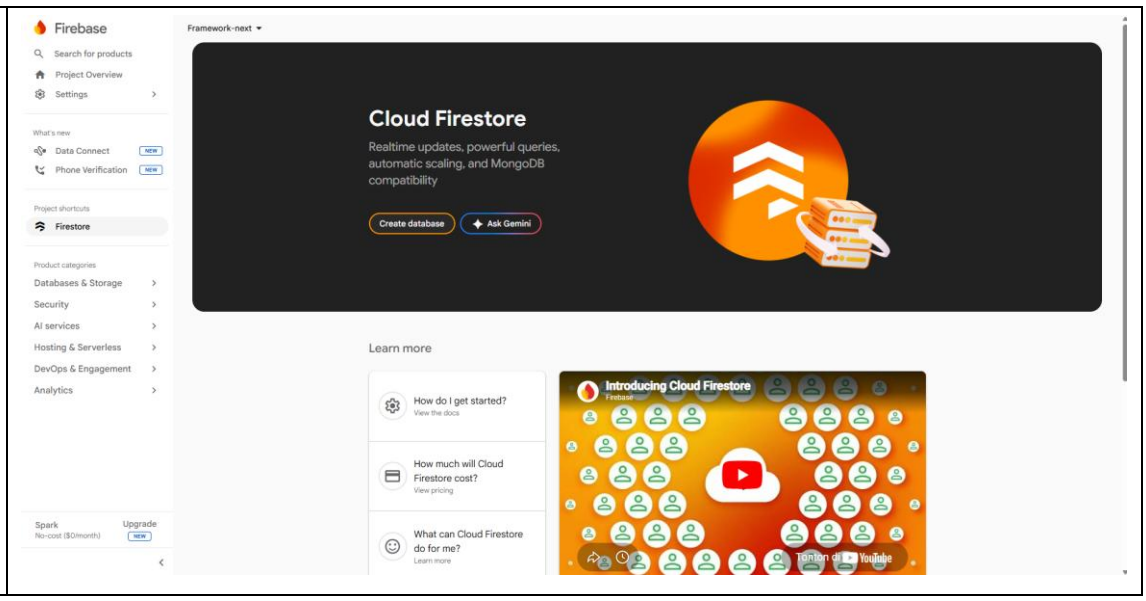

Create a project
Google Analytics
for your Firebase project
Google Analytics is a free and unlimited analytics solution that enables targeting, reporting, and more in Firebase Crashlytics, Cloud Messaging, In-App Messaging, Remote Config, A/B Testing, and Cloud Functions.
Google Analytics enables:
A/B testing
User segmentation & targeting across Firebase products
Breadcrumb logs in Crashlytics
Event-based Cloud Functions triggers
Free unlimited reporting
Enable Google Analytics for this project
Recommended
Previous
Create project

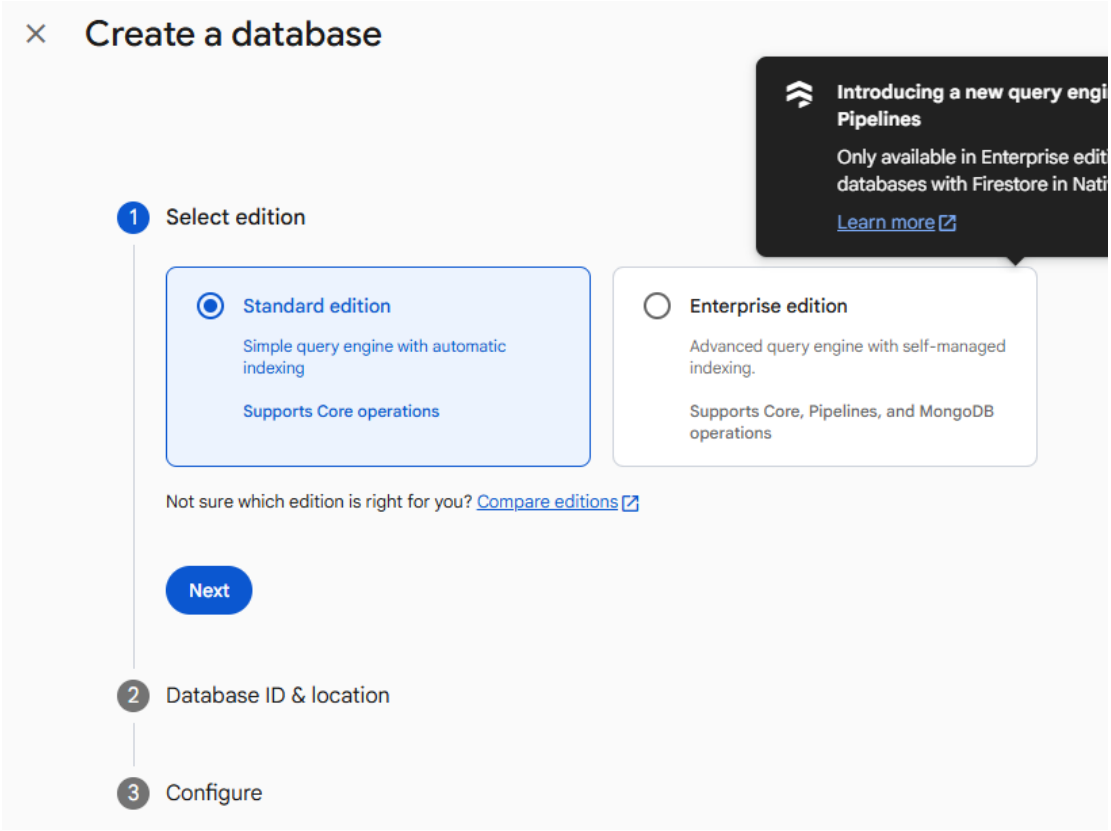

Framework-next
Your Firebase project is ready
Continue

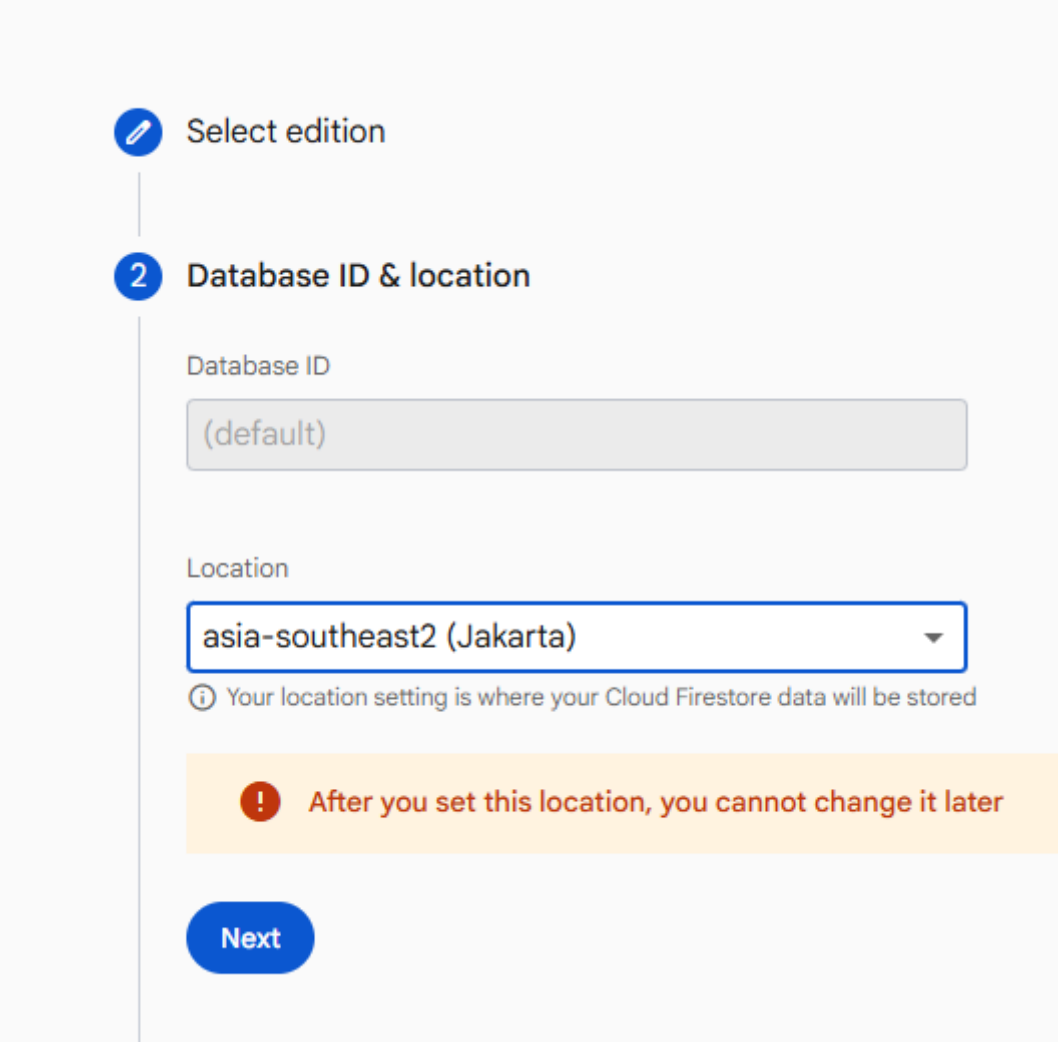

Framework-next
Spark plan
Select a platform
Hello, dimas adi bayu samudra
Welcome to your Firebase project!
Gemini can give you suggestions for Firebase services and solutions for your app
Next steps with Gemini
View docs
Tell us about your app
Describe your app, and Gemini will suggest products to help you get started
Add Analytics and monitoring
Learn more about Google Analytics and Firebase monitoring products
Build a backend
Firebase offers many backend services, including both SQL and NoSQL database options. Learn more about which services might be right for your app.
Add AI to your app
Learn more about integrating AI into your app
Monetize your app
Learn more about AdMob, personalizing user experience, and increasing revenue
Host a web app
Build and deploy modern, full-stack web apps

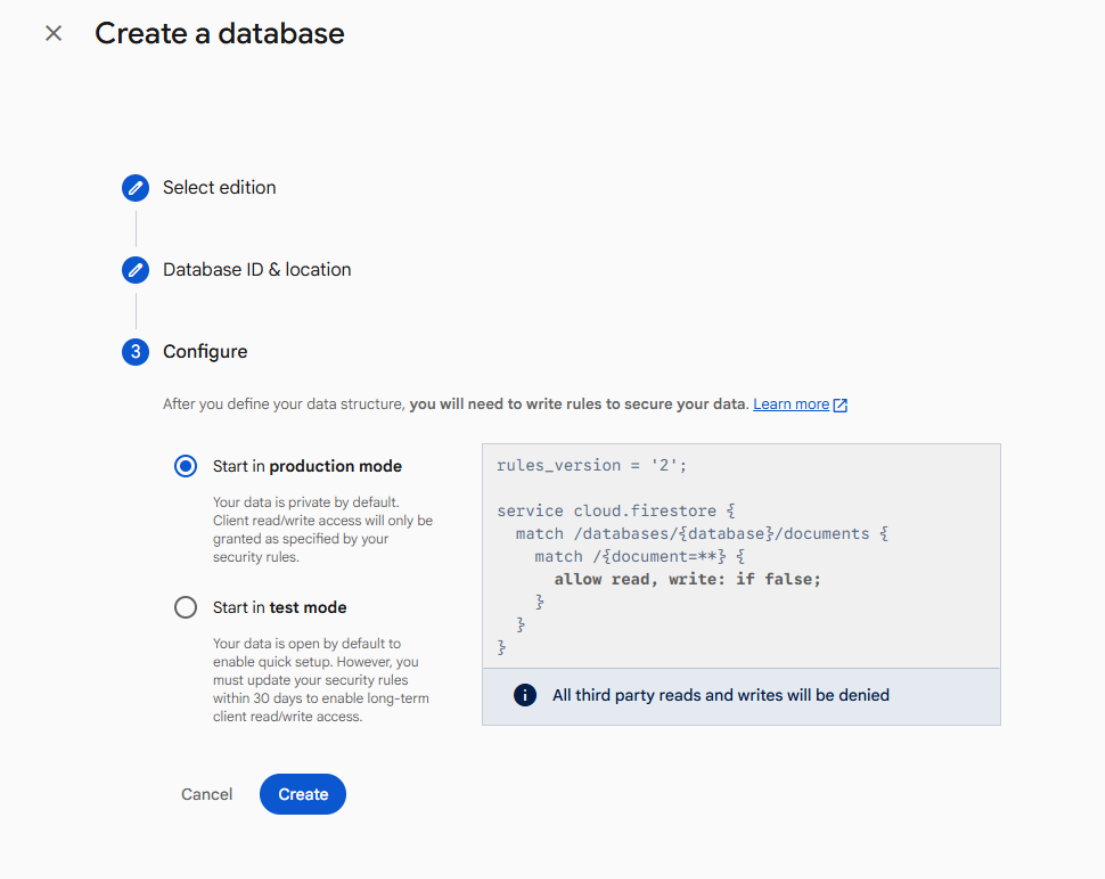

Add Firebase to your web app
1 Register app
App nickname
Framework-next2026
Also set up Firebase Hosting for this app. Learn more
Hosting can also be set up later. There is no cost to get started anytime.
Register app
2 Add Firebase SDK

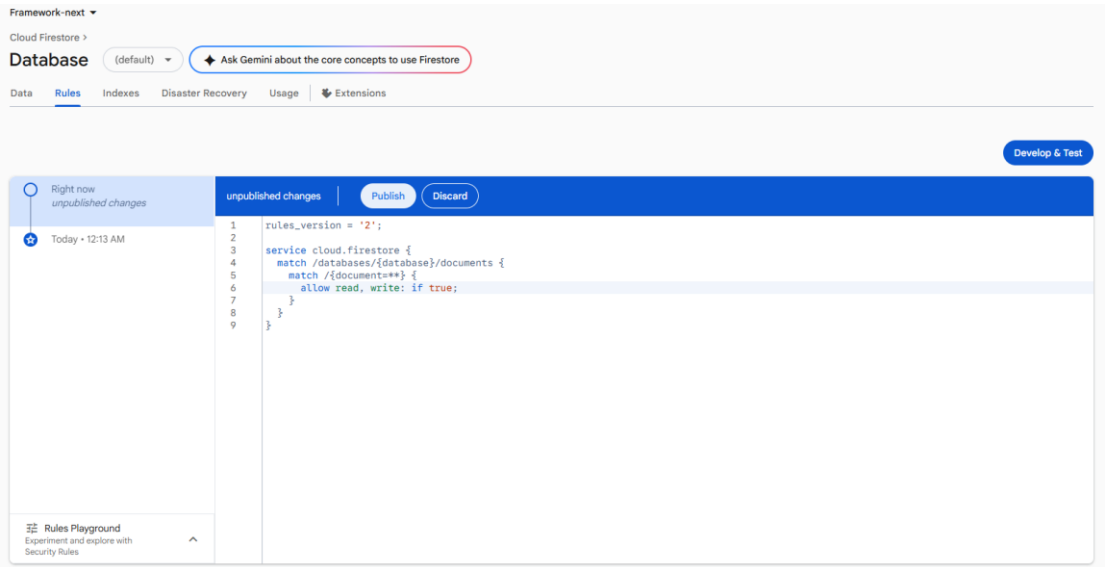
| | |
|---|---|
| |  <br>Register app<br>2 Add Firebase SDK<br>Use npm / Use a <script> tag<br>If you're already using npm and a module bundler such as webpack or Rollup, you can run the following command to install the latest SDK (Learn more):<br>`$ npm install firebase`<br>Then, initialize Firebase and begin using the SDKs for the products you'd like to use.<br>`// Import the functions you need from the SDKs you need`<br>`import { initializeApp } from "firebase/app";`<br>`// TODO: Add SDKs for Firebase products that you want to use`<br>`// https://firebase.google.com/docs/web/setup#available-libraries`<br>`// Your web app's Firebase configuration`<br>`const firebaseConfig = {`<br>`apiKey: "AIzaSyDirAtuI7FurQlbla-tLOy9nsCg1Mu3UYo",`<br>`authDomain: "framework-next-f25df.firebaseapp.com",`<br>`projectId: "framework-next-f25df",`<br>`storageBucket: "framework-next-f25df.firebasestorage.app",`<br>`messagingSenderId: "940134867545",`<br>`appId: "1:940134867545:web:c428e16b56678ee0758748"`<br>`};`<br>`// Initialize Firebase`<br>`const app = initializeApp(firebaseConfig);`<br>**Note:** This option uses the modular JavaScript SDK, which provides reduced SDK size.<br>Learn more about Firebase for web: Get Started, Web SDK API Reference, Samples<br>Continue to console |
| 3 | Aktifkan Firestore Database |

| | |
|---|---|
| |  Firebase<br>Search for products<br>Project Overview<br>Settings<br>What's new<br>Data Connect<br>Phone Verification<br>Project shortcuts<br>Firestore<br>Product categories<br>Databases & Storage<br>Security<br>AI services<br>Hosting & Serverless<br>DevOps & Engagement<br>Analytics<br>Spark<br>Upgrade<br>Framework-next<br>Cloud Firestore<br>Realtime updates, powerful queries, automatic scaling, and MongoDB compatibility<br>Create database<br>Ask Gemini<br>Learn more<br>How do I get started?<br>How much will Cloud Firestore cost?<br>What can Cloud Firestore do for me?<br>Introducing Cloud Firestore<br>YouTube |

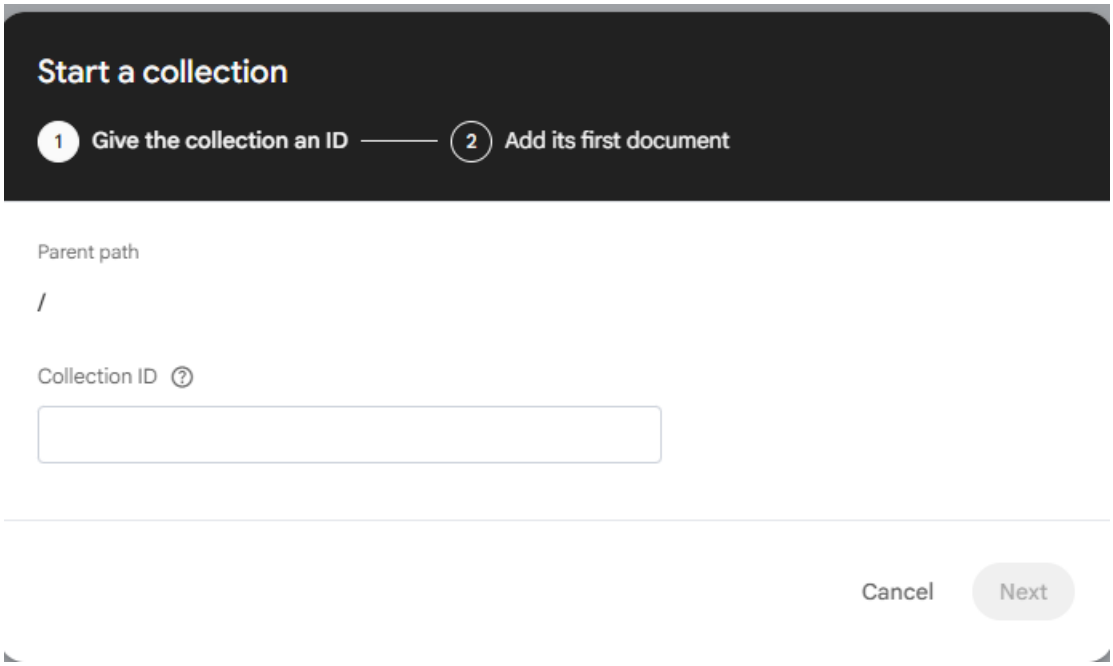
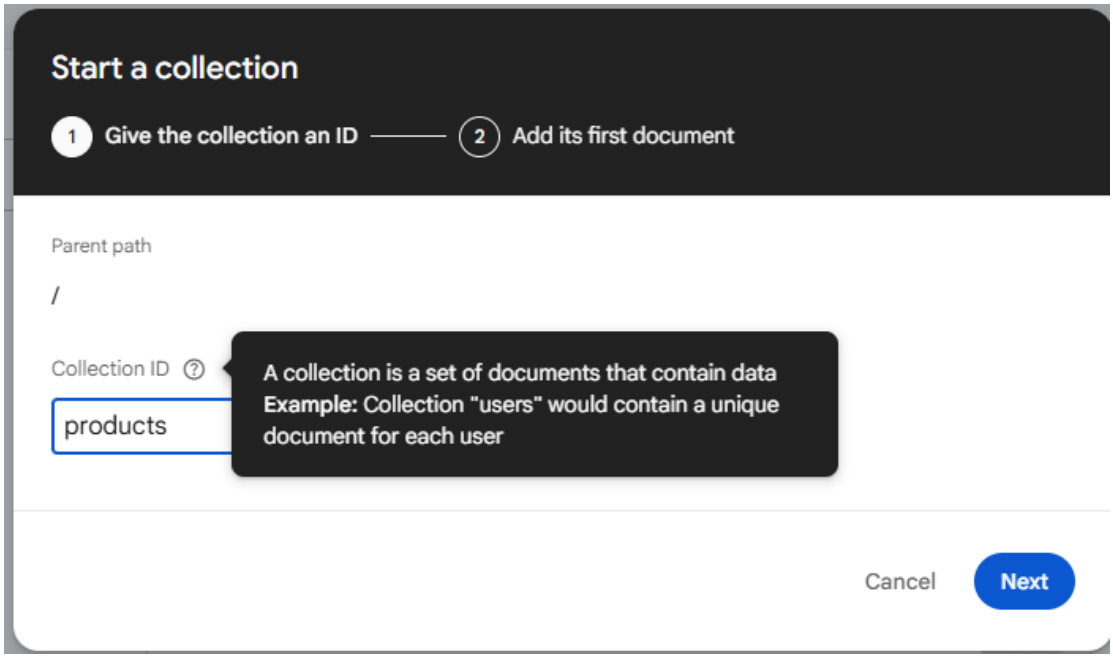

Create a database
Introducing a new query engi
Pipelines
Only available in Enterprise edit
databases with Firestore in Nati
Learn more
1 Select edition
Standard edition
Simple query engine with automatic indexing
Supports Core operations
Enterprise edition
Advanced query engine with self-managed indexing.
Supports Core, Pipelines, and MongoDB operations
Not sure which edition is right for you? Compare editions
Next
2 Database ID & location
3 Configure

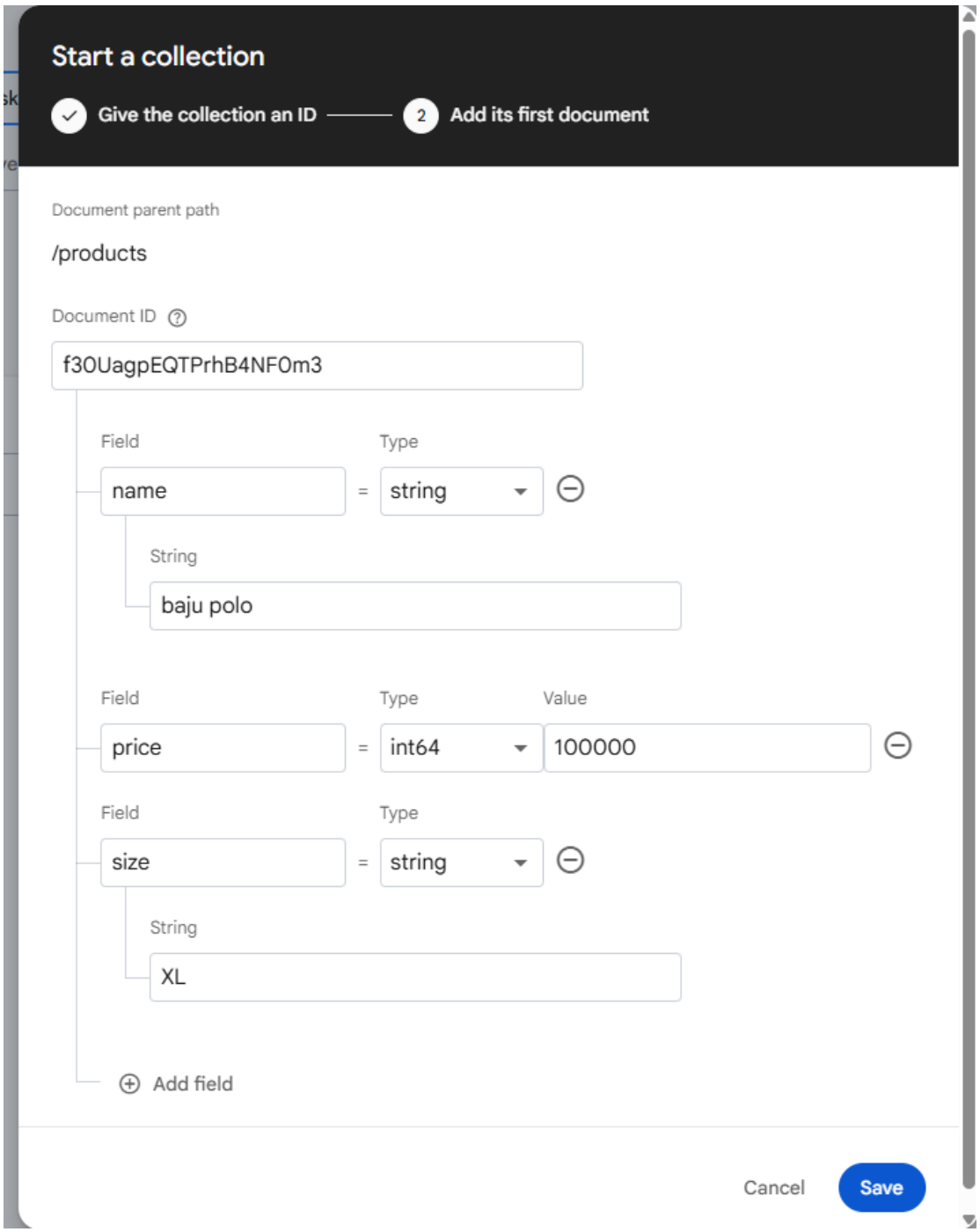

Select edition
2
Database ID & location
Database ID
(default)
Location
asia-southeast2 (Jakarta)
Your location setting is where your Cloud Firestore data will be stored
After you set this location, you cannot change it later
Next

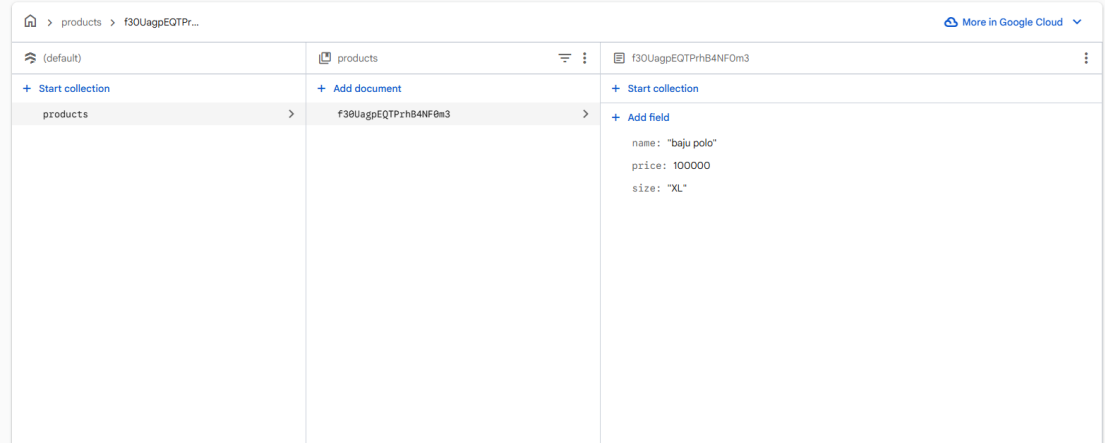


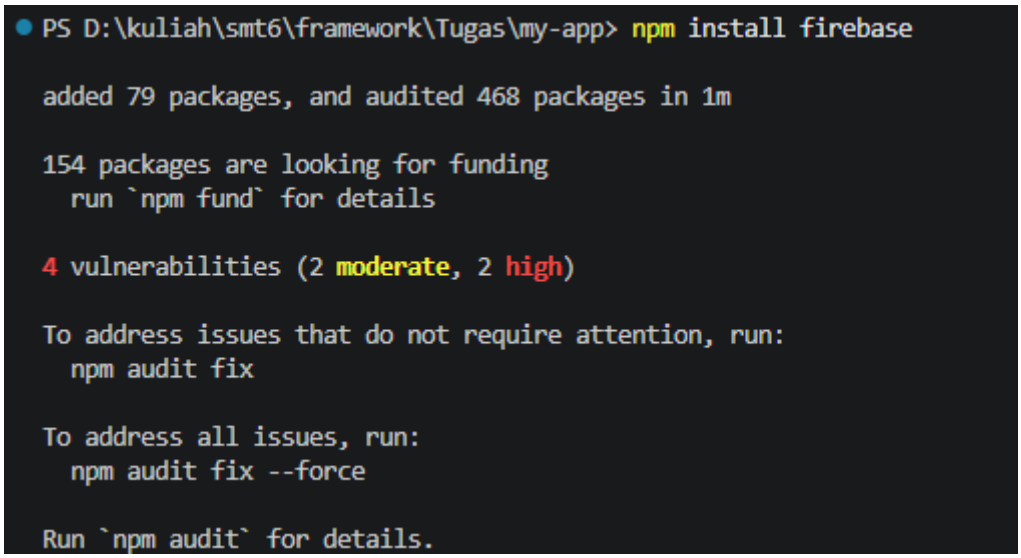
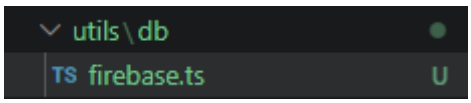

| | |
|---|---|
| 4 | Buat collection: |

| | |
|---|---|
| |  |
| 5 | Products<br>  |


Start a collection
Give the collection an ID
2
Add its first document
Document parent path
/products
Document ID
f30UagpEQTPrhB4NF0m3
Field
Type
name
=
string
String
baju polo
Field
Type
Value
price
=
int64
100000
Field
Type
size
=
string
String
XL
Add field
Cancel
Save

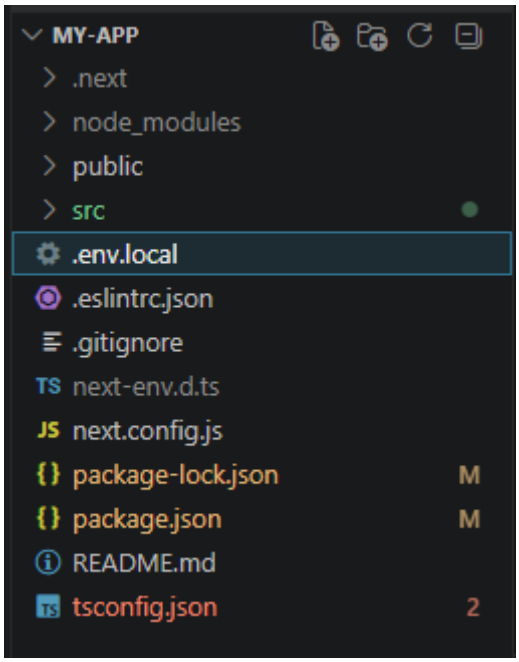
| | |
|---|---|
| |  |

## Praktikum Ke-5

| Langkah | Jawaban/Deskripsi |
|---|---|
| 1 | Install firebase<br>  |
| 2 | Buat folder dan file ts pada pages utlis/db/firebase.ts<br>  |
| 3 | Copy paste yang ada pada kotak merah ke file firebase.ts |

| | |
|---|---|
| | src > pages > utils > db > TS firebase.ts > ...<br>`// Import the functions you need from the SDKs you need`<br>`import { initializeApp } from "firebase/app";`<br>`// TODO: Add SDKs for Firebase products that you want to use`<br>`// https://firebase.google.com/docs/web/setup#available-libraries`<br><br>`// Your web app's Firebase configuration`<br>`const firebaseConfig = {`<br>`  apiKey: "AIzaSyDirAtuI7FurQlbla-tLOy9nsCg1Mu3UYo",`<br>`  authDomain: "framework-next-f25df.firebaseapp.com",`<br>`  projectId: "framework-next-f25df",`<br>`  storageBucket: "framework-next-f25df.firebasestorage.app",`<br>`  messagingSenderId: "940134867545",`<br>`  appId: "1:940134867545:web:c428e16b56678ee0758748"`<br>`};`<br><br>`// Initialize Firebase`<br>`const app = initializeApp(firebaseConfig);` |

**Praktikum Ke-6**

| Langkah | Jawaban/Deskripsi |
|---|---|
| 1 | Buat file: .env.local<br>  <br> |

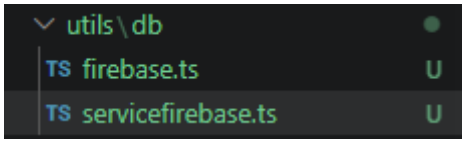
| | |
|---|---|
| 2 | Modifikasi file env<br>.env.local<br>FIREBASE_API_KEY=AIzaSyDirAtuI7FurQlbla-tLOy9nsCg1Mu3UYo,<br>FIREBASE_AUTH_DOMAIN=framework-next-f25df.firebaseapp.com,<br>FIREBASE_PROJECT_ID=framework-next-f25df,<br>FIREBASE_STORAGE_BUCKET=framework-next-f25df.firebasestorage.app,<br>FIREBASE_MESSAGING_SENDER_ID=940134867545,<br>FIREBASE_APP_ID=1:940134867545:web:c428e16b56678ee0758748 |

**Praktikum Ke-7**

| Langkah | Jawaban/Deskripsi |
|---|---|
| 1 | Modifikasi firebase.ts<br>src > pages > utils > db > TS firebase.ts > default<br>// Import the functions you need from the SDKs you need<br>import { initializeApp } from "firebase/app";<br>// TODO: Add SDKs for Firebase products that you want to use<br>// https://firebase.google.com/docs/web/setup#available-libraries<br><br>// Your web app's Firebase configuration<br>const firebaseConfig = {<br>apiKey: "AIzaSyDirAtuI7FurQlbla-tLOy9nsCg1Mu3UYo",<br>authDomain: "framework-next-f25df.firebaseapp.com",<br>projectId: "framework-next-f25df",<br>storageBucket: "framework-next-f25df.firebasestorage.app",<br>messagingSenderId: "940134867545",<br>appId: "1:940134867545:web:c428e16b56678ee0758748"<br>};<br><br>// Initialize Firebase<br>const app = initializeApp(firebaseConfig);<br><br>export default app; |

## Praktikum Ke-8

| Langkah | Jawaban/Deskripsi |
|---|---|
| 1 | Buat file<br>- ulits/db/servicefirebase.js<br>- modifikasi file servicefirebase.js |




```
src > pages > utils > db > servicefirebase.ts > retrieveProducts > data
import { getFirestore, collection, getDocs } from "firebase/firestore";
import app from "./firebase";

const db = getFirestore(app);

Windsurf: Refactor | Explain | Generate JSDoc | X
export async function retrieveProducts(collectionName: string){
    const snapshot = await getDocs(collection(db, collectionName));
    const data = snapshot.docs.map((doc) => ({
        id: doc.id,
        ...doc.data(),
    }));
    return data;
}
```

## Praktikum Ke-9

| Langkah | Jawaban/Deskripsi |
|---|---|
| 1 | Edit pages/api/product.js: |

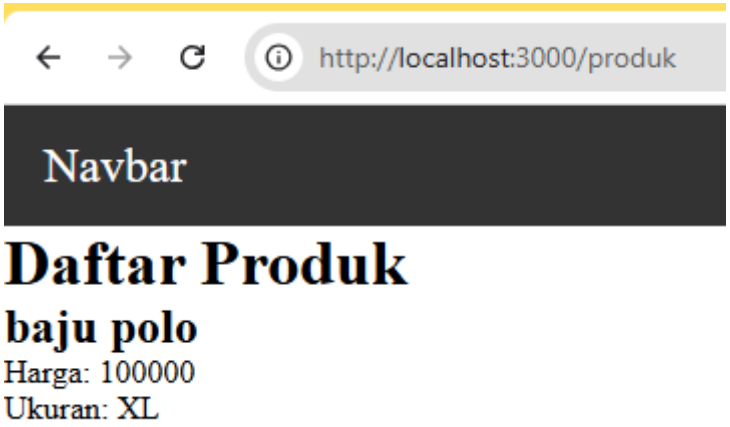
```
src > pages > api > produk.ts > handler
import type { NextApiRequest, NextApiResponse } from "next";
import { retrieveProducts } from "../utils/db/servicefirebase";

Windsurf: Refactor | Explain
type Data = {
    status: boolean;
    status_code : number;
    data: any;
};

Windsurf: Refactor | Explain | Generate JSDoc | X
export default async function handler(
    req: NextApiRequest,
    res: NextApiResponse<Data>
) {
    const data = await retrieveProducts("products");
    res.status(200).json({ status: true, status_code: 200, data });
}
```

| | |
|---|---|
| 2 | Jalankan browser http://localhost:3000/api/produk<br>http://localhost:3000/api/produk<br>Pretty-print<br>{"status":true,"status_code":200,"data":[{"id":"f30UagpEQTPrhB4NF0m3","price":100000,"name":"baju polo","size":"XL"}]} |
| 3 | Modifikasi index.ts pada produk sesuaikan nama typenya dan db nya |

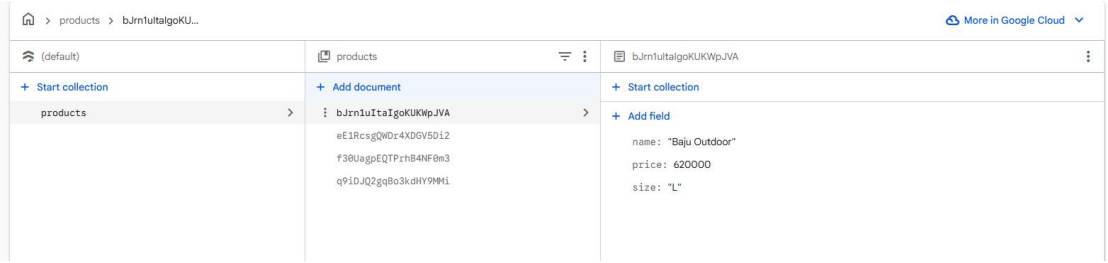
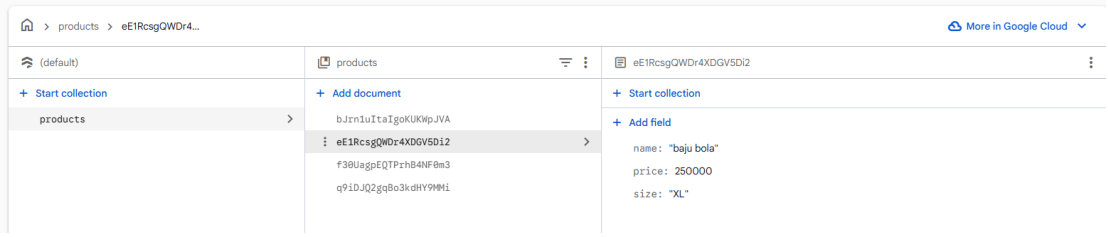
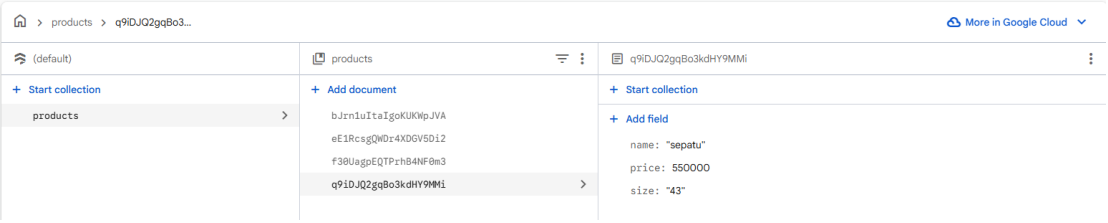
src > pages > produk > index.tsx > kategori > products.map() callback

You, 17 seconds ago | 1 author (You)

```
1  import ProdukPage from "../../views/produk";
2  import { useEffect, useState } from "react";
3
   Windsurf: Refactor | Explain
4  type ProductType = {
5      id: string;
6      nama: string;
7      price: number;
8      size: string;
9  };
10
   Windsurf: Refactor | Explain | Generate JSDoc | X
11 const kategori = () => {
12     const [products, setProducts] = useState<ProductType[]>([]);
13
14     useEffect(() => {
15   fetch("/api/produk")
16     .then((response) => response.json())
17     .then((responsedata) => {
18       //console.log("Data produk:", responsedata.data);
19       setProducts(responsedata.data);
20     })
21     .catch((error) => {
22       console.error("Error fetching produk:", error);
23     });
24 }, []);
25
26 return (
27   <div>
28     <h1>Daftar Produk</h1>
29     {products.map((product: ProductType) => (
30       <div key={product.id}>
31         <h2>{product.nama}</h2>
32         <p>Harga: {product.price}</p>
33         <p>Ukuran: {product.size}</p>      You, 6 seconds ago • Uncommitted
34       </div>
35     ))}
36   </div>
37 );
38 };
```

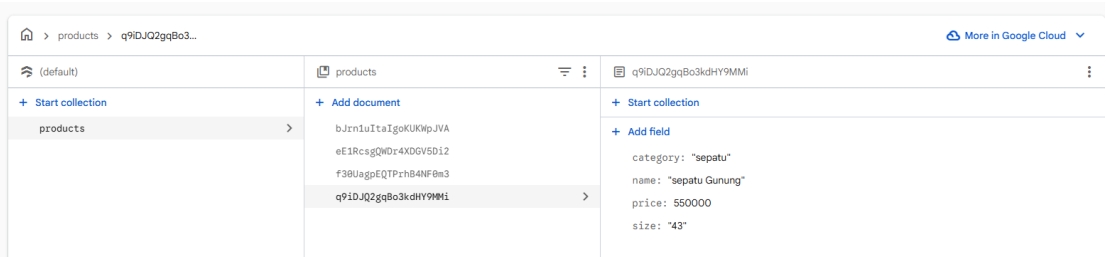
| | |
|---|---|
| 4 | Jalankan browser<br>  <br> |

**Tugas 1**

| Langkah | Jawaban/Deskripsi |
|---|---|
| 1 | • Tambahkan minimal 3 data produk di Firestore<br>• Pastikan data tampil di halaman produk<br>  <br><br>  <br><br>  <br> |

| | |
|---|---|
| | |

**Tugas 2**

| Langkah | Jawaban/Deskripsi |
|---|---|
| 1 | • Tambahkan field baru:<br>- category<br>• Tampilkan category di frontend<br>  |

```
Windsurf: Refactor | Explain
type ProductType = {
    id: string;
    name: string;
    price: number;
    size: string;
    category: string;
};

Windsurf: Refactor | Explain | Generate JSDoc | X
const kategori = () => {
    const [products, setProducts] = useState<ProductType[]>([]);

    useEffect(() => {
  fetch("/api/produk")
    .then((response) => response.json())
    .then((responsedata) => {
      //console.log("Data produk:", responsedata.data);
      setProducts(responsedata.data);
    })
    .catch((error) => {
      console.error("Error fetching produk:", error);
    });
}, []);

return (
  <div>
    <h1>Daftar Produk</h1>
    {products.map((product: ProductType) => (
      <div key={product.id}>
        <h2>{product.name}</h2>
        <p>Harga: {product.price}</p>
        <p>Ukuran: {product.size}</p>
        <p>Kategori: {product.category}</p>        You, now • Uncomm
```

http://localhost:3000/produk

Navbar

# Daftar Produk

## Baju Outdoor
Harga: 620000
Ukuran: L
Kategori: baju

## baju bola
Harga: 250000
Ukuran: XL
Kategori: baju

## baju polo
Harga: 100000
Ukuran: XL
Kategori: baju

## sepatu Gunung
Harga: 550000
Ukuran: 43
Kategori: sepatu

**Tugas 3**

| Langkah | Jawaban/Deskripsi |
|---|---|
| 1 | • Tambahkan tombol Refresh Data<br>• Gunakan fetch ulang tanpa reload halaman<br> <br> |

**Pertanyaan**

| Langkah | Jawaban/Deskripsi |
|---|---|
| 1 | Apa fungsi API Routes pada Next.js?<br>- memungkinkan Next.js bekerja sebagai backend , kita bisa membuat titik akses data (endpoint) di dalam folder pages/api/ yang menghasilkan respons berupa data (format JSON), bukan tampilan halaman web. |

| | |
|---|---|
| 2 | Mengapa .env.local tidak boleh di-push ke repository?<br>- File .env.local tidak boleh di-push ke repository karena berisi kredensial atau data rahasia (seperti API Key Firebase). Jika di-push ke repository (terutama yang bersifat publik), orang lain bisa melihat dan menyalahgunakan kunci akses tersebut untuk mengakses database. |
| 3 | Apa perbedaan data statis dan data dinamis?<br>- Data Statis: Data yang ditulis langsung di dalam kode program (hardcoded) dan tidak berubah kecuali kodenya diubah manual.<br>- Data Dinamis: Data yang diambil dari sumber luar, seperti database (Firebase), yang isinya bisa berubah-ubah sewaktu-waktu tanpa perlu mengubah kode program |
| 4 | Mengapa Next.js disebut framework fullstack?<br>- Next.js disebut framework fullstack karena ia memiliki kemampuan mengelola sisi tampilan (frontend) sekaligus sisi server atau logika database (backend) dalam satu project yang sama. Anda bisa membuat antarmuka pengguna sekaligus membuat API untuk mengolah data tanpa memerlukan server backend terpisah |